Agriculture, Ecosystems and Environment 183 (2014) 185–196

Contents lists available at ScienceDirect

Agriculture, Ecosystems and Environment

journal homepage: www.elsevier.com/locate/agee

Effects of trees on infiltrability and preferential flow in two contrasting agroecosystems in Central America

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ARTICLE INFO

Article history:
Received 15 April 2013
Received in revised form 18 October 2013
Accepted 25 October 2013
Available online 8 December 2013

Keywords:
Rainfall intensities
Trees outside the forest
Soil–water physical properties
Macroporosity
Tropics

ABSTRACT

We tested the hypothesis that trees have measurable effects on infiltrability, macroporosity, and preferential flows in agrosilvopastoral systems. Managing agricultural systems for water conservation is a critical component of sustainable systems. We investigated the relationship between infiltrability and the distance to the nearest tree, and whether differences in macroporosity can account for differences in infiltrability.

In both systems, preferential soil water flows were dominant compared to matrix flow. Trees in the pasture landscape improved infiltrability and preferential flow but had no significant effect in the coffee agroforestry system. After comparing rainfall intensity and frequency data to the measured infiltrability values, we conclude that trees in the pasture system reduce surface runoff at the highest observed rainfall intensities (>50 mm h^{-1}). The volcanic soils of the coffee plantation are less degraded and their high natural permeability has been maintained. Since the coffee plants at this site are established (40 years) perennial vegetation with substantial residues and extensive root systems like trees, they improve soil physical properties similarly to trees.

Trees increase hydrologic services in pasture lands, a rapidly expanding land use type across Latin America, and therefore may be a viable land management option for mitigating some of the negative environmental impacts associated with land clearing and animal husbandry. However, in land management practices where understorey perennial vegetation makes up a large proportion of the cover, such as for coffee agroforestry systems, the effect of trees on infiltration-related ecosystem services could be less pronounced

1. Introduction

Cattle farming has been the most rapidly-growing land use type in Central America, with the area of land allocated to this purpose increasing by 7 million hectares during the period of 1980–2000 (Gibbs et al., 2010). According to the FAO, there were 85 million hectares of permanent meadows and pasture lands in Central America in 2009, representing 35% of the region's usable land (FAO, 2012). It has been predicted that the establishment of new pastures will be the driving force for 69% of the deforestation that is projected to occur from 2010 onwards in this region (Wassenaar et al., 2007). Because of this expansion of pastures, it has been argued that it will be necessary to transform the means of livestock production in Latin America (Murgueito and Ibrahim, 2001). Methods based around increasing the tree cover provided by conventional grass monocultures could play a central role in this transformation (Murgueitio et al., 2011).

Water quality and quantity are important limiting factors that affect socioeconomic development, human health and environmental sustainability in many regions of the world (Meadows and Meadows, 2007; Meadows et al., 1974). In recent years, the concept of ecosystem services has been used to integrate scientific understanding of biophysical processes with socio-economic analysis. The storage and retention of water, and the regulation of water supplies, have been described as hydrological ecosystem services (HES; Costanza et al., 1997; de Groot et al., 2002).

Facing the rapid expansion of pasturelands across Latin America, we need strategies that can easily be implemented on farms to reduce environmental impacts and improve water quality and quantity in particular, where trees may provide such services, but data are sparse.

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http://dx.doi.org/10.1016/j.agee.2013.10.027

Previous studies on the relationship between trees and water distribution focused primarily on forest plantations (Bruijnzeel, 2004; Scott et al., 2005; van Dijk and Keenan, 2007); the effects of isolated trees or small clusters of trees in livestock farms and agroforestry systems on groundwater recharge have not been studied extensively. However, one recent study in this area found that the level of water infiltration in a coffee agroforestry system was greater than that in a coffee monoculture (Cannavo et al., 2011).

Trees modify the microclimate (Charbonnier et al., 2013) and variables that affect the water balance at the local scale, such as interception, transpiration, infiltration, surface runoff and soil evaporation. Depending on the circumstances, these effects may or may not be beneficial in terms of maintaining adequate ground water recharge rates and dry season flows (Bruijnzeel, 2004). Over the last few years, numerous afforestation and reforestation projects have been initiated with the aim of mitigating the effects of climate change. An underappreciated issue associated with such projects is their potential effects on hydrological ecosystem services, which arise from the trees' high water use reducing downstream water flow (Trabucco et al., 2008). A recent meta-analysis showed that the total flow levels for planted and natural forests are lower than those for non-forest lands (Locatelli and Vignola, 2009). Jackson et al. (2005) reported that tree plantations decreased both annual total and dry season stream flows. However, it has been pointed out that these studies focused exclusively on data for subhumid tropical and temperate areas (Malmer et al., 2010). This is significant because the soil water balance is dependent on soil mineralogy and texture, local rainfall patterns, and tree cover across the landscape. At the local scale, trees affect flow in two main ways. First, they influence the soil's permeability and thus affect positively the soil water infiltration. Second, they increase evapotranspiration because of canopy interception and the uptake of water in the root zone via the roots. There is also evidence that forests affect rainfall patterns at the regional level (Ellison et al., 2012; Makarieva et al., 2013; Makarieva and Gorshkov, 2007). Even over timescales of less than a day, evapotranspiration in the cover can be expected to influence the streamflow (Bond et al., 2002; Cadol et al., 2012).Trees often have deep root systems and produce more residues than other cover types, increasing the soil organic matter content and improving water retention in the top soil. This also increases the soil's porosity and facilitates water infiltration (Ilstedt et al., 2007). While meta-analyses have shown that the introduction of trees into agricultural fields and afforestation in the tropics can both increase infiltrability, quantified by the rate of water infiltrated in the soil until reaching a steady state ratio, in general, Ilstedt et al. (2007) have stressed that the existing data are insufficient to fully understand the precise impacts of such changes at the local level. For example, there may be situations in which trees increase infiltration to a much lesser extent (or do not increase it at all) than would be expected based on the general case. The inherent macroporosity of naturally permeable soils such as those of volcanic origin may be so high that tree roots and litter do not increase it significantly or contribute meaningfully to soil infiltrability. In addition, the understory may also affect infiltration positively or negatively, and depending on its nature (e.g. whether it is deep rooted or not, perennial or not, compacted or not) may also interact with the overstory trees. It is therefore important to characterize different situations in order to determine how trees within the agroecosystem affect the soil water balance. The process of infiltration affects the magnitude and quality of both soil- and ground-water systems (Wu et al., 1996), and is essential for their replenishment and maintenance (Grinevskii and Novoselova, 2011). Infiltration has therefore been studied for a long time, particularly in the context of agriculture (Chapman, 1990). However, there is still a lack of data on infiltration in various soil types found in the tropics, and how it is altered by land management in these regions (Bruijnzeel, 2004; Malmer et al., 2010). Simple hydrological process models often neglect the role of infiltrability and its dependence upon rainfall intensity, but there is an increasing awareness that both processes should be considered simultaneously (Jackson et al., 2005). This is particularly important in the tropics, which have very intense rainfall and where soils are often prone to degradation and compaction. In this region, the comparison of rainfall intensities with infiltrability is of main importance to evaluate the potential effects of these intense rainfalls on runoff generation and soil erosion. It is important to understand how trees affect infiltrability to facilitate the accurate parameterization of hydrological models in order to account for the spatial heterogeneity introduced by the presence of trees in the humid tropics (Ilstedt et al., 2007).

Trees are integral components of many cropping (e.g. coffee, cacao agroforestry) and silvopastoral systems in the tropics (Somarriba et al., 2012). Initiatives aimed at increasing the biodiversity within coffee-producing regions by introducing trees have been widely adopted in Central America, due to environmental concerns, together with low coffee prices (Muschler and Bonnemann, 1997). In addition, soil carbon sequestration and agroforestry are often promoted as management options in small-scale agriculture. This is driven by a multitude of factors, including the need for food security and to mitigate the impact of climate change (DeFries and Rosenzweig, 2010; Verchot et al., 2007).

To determine the role of isolated trees in the agrosilvopastoral systems of Central America, we investigated how trees affect soil infiltrability and macroporosity, which are two of the key variables that govern groundwater recharge. In this work we describe two agroecosystems that were selected to represent two extreme situations in terms of infiltration: an agroforestry-based coffee plantation located in the humid tropical environment of Costa Rica and a pasture landscape with an extended dry season in Honduras. The coffee agroforestry system was located on naturally permeable volcanic soils, and we assumed that the impact of the densely-rooted coffee plants on infiltration would be similar to those of trees. The pasture landscape had a lower level of tree cover, and we assumed that animal trampling would compact and degrade the topsoil. It was hypothesized that in both cases, the soil infiltrability and macroporosity would be influenced by (1) soil type, and (2) land management.

We therefore aimed to test the hypothesis that in agrosilvopastoral systems there is a positive effect of isolated trees on infiltration. Specifically, the aims of our study were to: (1) investigate the effects of isolated trees on infiltrability and preferential flows, and the distance over which these effects apply, (2) compare infiltrability and rainfall intensity, and (3) evaluate the importance of the prevailing conditions (soil type, land use patterns, understory composition).

2. Material and methods

2.1. Locations and sites

The aim of having these two study sites is not as a comparison, but to test the hypothesis that trees have a positive effect on infiltrability in these two systems that are common land uses in the tropics and where the inclusion of trees can have a role in hydrological ecosystem services.

2.1.1. Study site 1: Coffee plantation agroforestry system in Turrialba, Aquiares, Costa Rica

This study was conducted in a coffee agroforest located in the Aquiares farm, herein referred as Aquiares, one of the largest coffee farms in Costa Rica (6.6 km^2) and site of the Coffee-Flux observatory (SOERE F-ORE-T, http://www5.montpellier.inra.fr/ecosols/

Table 1
Texture, dry bulk density (DBD), and total porosity (TP) at two soil horizons, and soil organic matter (SOM) content of the superficial soil for both study sites. Standard deviations are given in parentheses.

Site/depth (cm)	Sand (%)	Clay (%)	Silt (%)	Dry bulk density-DBD ($g\,cm^{-3}$)	Total porosity-TP (%)	Number of samples for texture, DBD and TP	SOM (%)	Number of samples for SOM
Turrialba–Aquiares								
10 cm	58.46 (2.34)	14.16 (2.48)	27.37 (1.98)	0.71 (0.15)	73 (0.05)	12	13.26 (5.20)	60
100 cm	55.12 (6.57)	14.18 (4.46)	30.70 (3.38)	0.79 (0.15)	70 (0.06)	12	–	–
Copan–Honduras								
10 cm	45.15 (12.43)	28.88 (10.79)	25.9 (6.33)	0.68 (0.15)	74 (0.05)	12	10.18 (2.84)	71
100 cm	42.84 (16.09)	30.10 (16.55)	2.52 (6.77)	1.23 (0.16)	54 (0.06)	12	–	–

Recherche/Les-projets/CoffeeFlux). The farm is located within the subcatchment area of the Aquiares river on the slopes of the Aquiares Volcano, in the sub-humid tropics. The average annual precipitation at the farm between 1973 and 2010 have been 3014 mm, with the lowest rainfall period occurring between December and March. The farm has been certified by the Rainforest Alliance™ since 2003, in a program for shade-grown coffee aiming to conserve biodiversity. The experiments were conducted within the Mejias Creek microcatchment (−83°44′W and 9°56′N), which extends over an area of 0.9 km^2. The altitude of the experimental site ranges from 1020 to 1280 m.a.s.l, with a mean slope of 20%. The site has a number of permanent streams, with a drainage density of 6.2 km km^2. The Aquiares farm was homogeneously planted with coffee (*Coffea arabica* L., var Caturra) on bare soil, 40 years ago, with a density of 6300 plants ha^{-1}, shaded by large *Erythrina poeppigiana* trees, which have been allowed to grow freely since 2001 (Gómez-Delgado et al., 2011). The shade trees have a mean density of 10 trees ha^{-1} (15.7 ± SD 5.5% SD crown projected area) and a canopy height of ca. 20 m (Charbonnier et al., 2013). The soils are Eutrandepts Andisols (USDA, 1999) formed from volcanic ejecta by various weathering and mineral transformation processes. Soils of this type are very stable, with high contents of organic matter and levels of biological activity, high allophane contents (Kinoshita, 2012; Tobón et al., 2010), and layers with coarse textures (Table 1). The roots of the coffee plants and trees at this site were found to extend down to the bedrock, which is located at a depth of around 4 m (Defrenet, 2012). The site's watershed has very high infiltration rates (92% of rainfall; Gómez-Delgado et al., 2011). Infiltration is further facilitated by the presence of densely rooted tree crops like coffee, the absence of tillage, and the soil's high macroporosity (Dorel et al., 2000).

2.1.2. Study site 2: Pasture landscapes in the Copan River catchment area, Honduras

The Copan River drains into the Motagua River and has a transboundary catchment area of around 620 km^2 that extends from 14°43′ to 14°58′N and from 88°53′ to 89°14′W, covering parts of both Honduras and Guatemala. Pastures are the most common land use type in this region, accounting for 30% of the catchment's total area (MESAP-MANCORSARIC, FOCUENCAS II., 2006). The soils are Typic Argiustolls class, and lie above intrusive rocks and are generally deep with good drainage (USDA, 1987; Table 1). The region is a tropical savannah with a pronounced dry season that runs from December to May. The average annual precipitation is 1772 mm (Lopez, 2011, pers. comm, mean over 32 years).

We studied three farms located in the communities of El Malcote, Sesesmiles and El Zapote, all of which lie within the lower subcatchment of the Copan River. All of the studied farmlands have been used as cattle pastures for the last twenty years, and are covered with a mixture of native grass and introduced species (*Cynodon dactylon, Cynodon plectostachyus, Panicum maximum* and *Brachiaria brizantha*). Old native clumps of trees are sparsely distributed in and around the pasture. Based on surveys of farmers' opinions conducted as part of an ongoing project (CATIE-MESOTERRA, 2009), it was estimated that the farms have around 19 trees ha^{-1} (Benegas et al., 2011). However, the sites examined in this work contained only between 5 and 7 clumps of trees ha^{-1} (based on Google Earth images acquired in 2007). These trees generally have wide crowns, indicating that they have been growing for long periods of time in an open landscape. Conversely, some pastures located up-slope of the studied farms contain trees with narrow crowns that evidently grew up in forests. More recently-introduced woody components include species that are used as live fences to define farm boundaries, notably *Gliricidia sepium*. We examined the physical properties of the soil below and around clumps of *Byrsonima crassifolia, Lonchocarpus hedyosmus, Spondias purpurea, Ficus colibrinae and Lippia sp.* The mean animal load in Sesesmiles was 1.8 UA ha^{-1}, slightly higher than the sustainable livestock management level (1 to 1.3 UA ha^{-1}; Cristobal Villanueva, pers. comm). We estimate that the animal loads on the two other farms examined in this work (El Malcote and El Zapote) may be even greater. In all three cases, we observed signs of soil degradation due to trampling, and subsequent erosion.

2.2. Experimental design

At each site, we examined a series of transects that had one tree or a clump of trees at one end and no other trees along their length. In the coffee agroforest, the transects, originated at a single tree, contained a tree and coffee plants, with coffee plants only elsewhere along their length. In the pastures, the transects contained a clump of trees at the point of origin and open-pasture elsewhere along their length. All transects had their points of origin in the center of a tree (at Aquiares, Fig. 1a) or clump of trees (at Copan, Fig. 1b). Infiltrability tests were conducted along a line that also originated at the tree or clump of trees, and extended outwards in a randomly-chosen direction (north, west, south, or east). Six transects were established in the agroforest, and their locations stratified to represent the ranges of slopes and distances to stream. These transects were divided into two sections: (1) the section beneath the origin-canopy (under the tree crown) and (2) the section distant from the tree (under coffee plants only). We performed five double ring steady state infiltrability (DRSSI) tests at 1 m intervals. Therefore, a total of 60 infiltrability tests were conducted at Aquiares (Fig. 1a). For transects examined in the pasture, we took the center of the clump of trees as the point of origin, and, each transect was divided into an open-pasture section and the section beneath the canopy of the trees. Five infiltrability measurements were collected under the canopies of each transect, at 1 m intervals. In addition, we took two infiltrability measurements at three locations in the distant open-pasture sections of each transect. The positions of the open-pasture measurement sites were separated by a distance equal to the height of the tallest tree in the

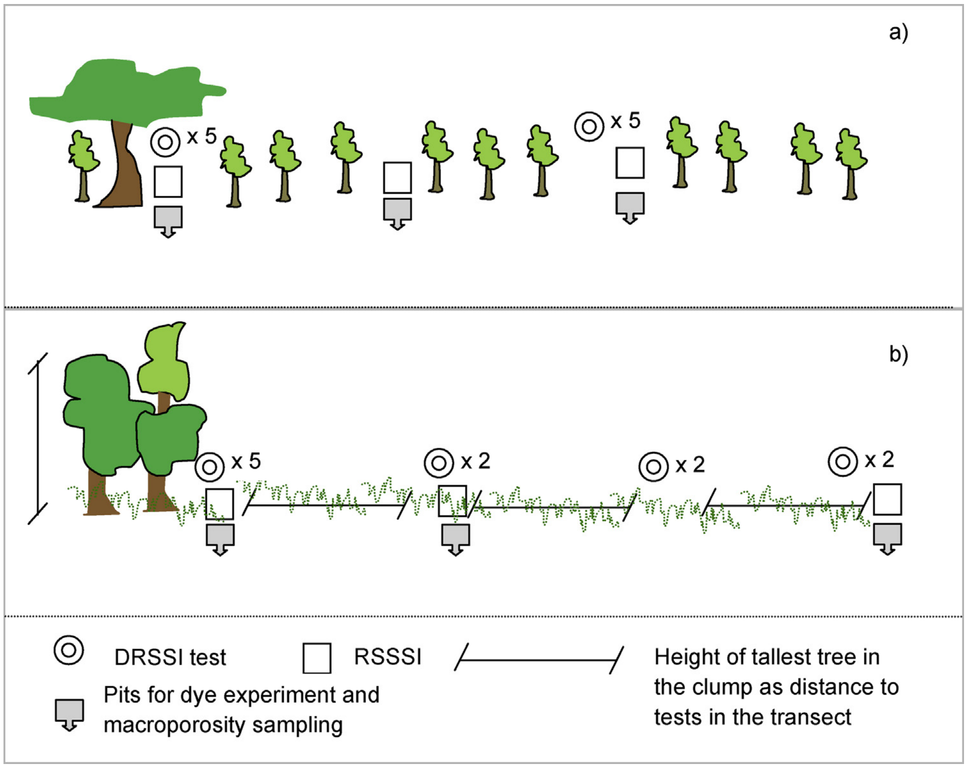

Fig. 1. Sketch of transects and main tests conducted in a) the coffee agroforestry system (Aquiares), and b) the pastoral system (Copan). DRSSI stands for double ring steady state infiltrability; RSSSI stands for rainfall simulation steady state infiltrability.

clump (see Fig. 1b). A total of seven transects were established, with five measurements beneath the canopies and six measurements in the open-pasture, per transect, 77 infiltrability measurements in Copan.

2.3. Rainfall intensity, infiltrability and superficial runoff

Rainfall intensity was measured at resolutions of 10 min and 5 min at Aquiares and Copan, respectively, using automatic rain gauges (Texas Electronics-TE525 series at Copan; Campbell Scientific ARG100 pluviometers at Aquiares) connected to dataloggers (Campbell Scientific CR800 series). Rainfall intensity data covered a period of 2.5 years (2009–2011) in Aquiares, and a period of 1.5 years (2010–2011) in Copan. The percentage of total rainfall for the corresponding study periods was calculated by summing the magnitudes of all rainfall events (in mm) at 5- and 10-min intervals. Reference intensities of 20, 30, 40, and 50 mm h^{-1} were used to determine the relationship between rainfall intensity and infiltrability.

Two complementary methods were used to measure infiltrability, (1) the double ring method for steady state infiltrability (DRSSI), and (2) rainfall simulation steady state infiltrability tests (RSSSI). The first method was used to obtain an estimate of the maximum steady state infiltrability (25 cm inner ring, 35 cm outer ring, 10 cm water head; Reynolds et al., 2002). Experiments were allowed to run until a steady rate of infiltration was achieved, and the data obtained were modeled using Philip's equation (Philip, 1957).This method gives a rough indication of the maximum ponded infiltrability that can be compared to rainfall amounts and intensities (Malmer and Grip, 1990). Such measurements are simple and inexpensive. However, this approach has two notable drawbacks: (1) the installation of the two rings disturbs the soil, affecting the results obtained, and (2) the ponding of the water increases the water pressure head relative to that which occurs during a normal rainfall event.

We also measured infiltrability with rainfall simulation steady state infiltrability tests (RSSSI) using pre-selected simulated rainfall intensities representative of the maximum in each region. The surface runoff volume was collected and infiltrability was measured indirectly with this method. The rainfall simulator had a water-emitting surface area of 60 cm^2, with water droplets emitted through nozzles having a diameter of 0.50 mm each. The simulator was mechanically oscillated during the experiments to ensure that the rainfall was evenly distributed within the experimental area, producing simulated rainfall intensities of around 70 mm h^{-1} on average, peaking at 80 mm h^{-1}. According to Jackson (1986) and van Dijk et al. (2002), (100 mm h^{-1}) is a plausible upper limit on the maximum rainfall intensity that will occur naturally in the tropics. The simulator used in this work was a modified version of that described by Imeson (1977). We used the maximum rainfall intensities registered in Copan at 5 min intervals to define the rainfall intensity simulated in rainfall simulation tests, i.e. 83 mm h^{-1}. All tests were conducted until a steady state of infiltration rate was achieved. The drawback of this approach compared to double ring steady state infiltrability (DRSSI) is its higher cost and greater time requirements. In addition, the maximum achievable simulated rainfall intensity imposes an upper limit on the range of steady state infiltrability values that can be achieved in that region. Rainfall simulation experiments were conducted at the point of origin of selected transects, as well as half way along their length and at the end of each transect; in total, 12 such experiments were performed at each study site (Fig. 1).

2.4. Soil properties, water holding capacity, preferential flow and macroporosity

At the end of each rainfall simulation steady state infiltrability test (RSSSI), we measured macropore flow using the dye method (Flury and Wai, 2003). A solution of 18 l of water and dye for coloring pores was prepared, containing 20 g of Brilliant Blue per liter (which corresponds to 300 mm of precipitation given the surface area of the rainfall simulator). After 1 h, the dye flow patterns in the soil were analyzed by excavation (see below for further details).

To determine the soil's water holding capacity (WHC), we sampled horizontally (through the vertical walls of the pits) undisturbed soil using stainless steel cylinder (diameter = 7.5 cm and height = 5 cm). The soil samples were weighed (field weight soil; FWS), then saturated from below by placing a filter tissue at the bottom of the cylinder (held in place with rubber rings) and slowly adding water every hour for 8 h until saturation was achieved, at which point the sample was re-weighed to determine the weight of the saturated soil (wet weight soil, WWS). The sample was drained for 1 h, and then re-weighed before and after drying to constant weight at 105 °C for 24 h, to obtain the dry weight soil (DWS). Thus, the equation used was WHC(%) = (WWS – DWS)/DWS

We define macroporosity as the abundance of pores with a diameter in excess of 0.05 mm. These are also known as transmission pores, and allow for rapid drainage after heavy rainfall or irrigation, and are important for soil aeration when the soil is at field capacity (Rowell, 1994). Macroporosity was assessed using the assumption that macropores are drained at WHC, and so the difference between total porosity and WHC provides a measure of macroporosity. The following expressions were used for this purpose (Rowell, 1994):

$$\text{Total porosity}\,(\%) = 1 - (D_b/D_p)$$

where

D_b = Soil bulk density (mass of soil dried to constant weight at 105 °C for 24 h, divided by cylinder volume: g cm^{-3})

D_p = soil particle density (assumed to be 2.65 g cm^{-3})

$$\text{Macroporosity}\,(\%) = \text{Total porosity}\,(\%) - \text{WHC}(\%)$$

Dye (Brilliant Blue) patterns were analyzed to determine whether there was a difference in macropore flow between the tree sections of the transects and the gaps. We dug a pit on one side of the runoff test areas and took photographs of the pit wall (which was located between the pit and the runoff area) at depth increments of 10 cm until no evidence of the dye. Pits were excavated to depths between 20 and 110 cm depending on the dye's penetration depth. We placed graduated frame on the pits' soil walls before photographing to provide references for image analysis. Digital photos were taken using a 24 mm wide angle lens (Sony Cyber-shot DSC-WX10), and the distance between the wall and the lens was 50 cm in all cases. Image analysis was performed using the ENVI image analysis program, version 4.7 (ITT Visual Information Solutions—www.ittvis.com) to determine the amount of dye within the soil shown in each photo-slice. We classified the two colors of interest (soil with and without Brilliant Blue) using a supervised Mahalanobis distance classification (Richards, 1999). In each photograph, the area defined by the frames was used as individual region of analysis. We then summed each frame depth until no dye was found, to account for the whole region of analysis in each pit. We related the number of pixels covered by the color of interest (blue) in the region of analysis with the soil without dye cover. Values were expressed as percentages of blue coverage at each depth and position (horizontal and vertical).

Contrasting examples are presented in Fig. 2, which shows some of the image analysis results obtained for the two study sites at various depths. The images were interpreted by comparing the percentage of stain coverage over the percentage of unstained soil. In this case, a greater extent of staining indicates a lower number of macropores capable of transporting water; the smaller the stained area, the greater the number of macropores within the soil and the more efficient the transport of water.

The dye cover images were used to calculate a number of preferential flow indexes: (1) the uniform infiltration depth, (2) the maximum depth of blue stains, (3) the total stained area, and (4) the preferential flow fraction. As described by van Schaik (2009), we assumed cases where ≥80% of the area in an image was stained to be indicative of uniform infiltration flow, which is associated with micropores (<0.2 μm) and mesopores (10–0.2 μm; Sander, 2002). Cases where the level of staining was below this threshold were assumed to be indicative of preferential flow due to abundant macropores. In addition to these variables, we calculated the total stained area as a function of the maximum depth of each pit as a new integrated index of preferential flow. This was done to account for the varied depths of the pits at each location. The methods used to calculate these preferential flow parameters are explained in Table 2.

2.5. Statistical analyses

To test for differences in infiltrability rates (as measured by DRSSI) in the vicinity of trees (near trees, up to 5 m apart from trees) and in the gaps between trees (distant trees, up to 30 m apart from trees), we used the paired *t*-test for the Aquiares dataset, which was normally distributed. The Copan dataset was non-normal and so in this case a median comparison was performed by means of the Mood test. These calculations were performed using Minitab 16 (Kowalski and Montgomery, 2010).

We also used principal component analysis (PCA) to explore the relationships among the following soil preferential flow parameters: rainfall simulation steady state infiltrability (RSSSI), macroporosity (MC), and the distance-to-trees (DTT). Only significant components were reported. The data were scaled to unit variance and mean centered to account for differences in scale and variation between variables. PCA was performed using the statistical software package SIMCA-P (UMETRIC, 2001), since this program is able to provide significance estimates for the components based on cross-validation, and to calculate confidence intervals for variable loadings using a Jack-knife error estimator (Efron and Gong, 1983).

3. Results

3.1. Infiltrability compared with rainfall intensity

3.1.1. Rainfall intensities

There was higher rainfall intensity in Copan compared to Aquiares, where heavy rainfall (>20 mm h^{-1}) had 30% likelihood in Copan, and only 12% likelihood in Aquiares. Fig. 3a shows the rainfall intensities over a period of 33 months (Sept. 25, 2009 through Dec. 31, 2011). The total rainfall during this period at Aquiares was 5700 mm, and the maximum 10-min intensity was 106 mm h^{-1}.

In contrast to Aquiares, the rainfall intensities at Copan were more heterogeneous, and heavy rainfall was more common (Fig. 3b). In this case, the maximum 5-min intensity for the period was 82.9 mm h^{-1}, and the maximum 10-min intensity was 41.4 mm h^{-1}. Rainfall levels were monitored over 20 months (May 18, 2010 to Dec. 31, 2011); the total precipitation during this time was 2600 mm.

No clear relationship between rainfall levels and time were identified for any rainfall event that occurred during this work. However, there were a few rainfall events of up to 50 mm h^{-1} that lasted for at least 20 min in Aquiares, and at least 10 min at Copan (Table 3). These downpours accounted for the majority of the rainfall during the events in which they occurred in the agroforest. Nevertheless, in the pasture, rainfall intensities of >50 mm h^{-1}

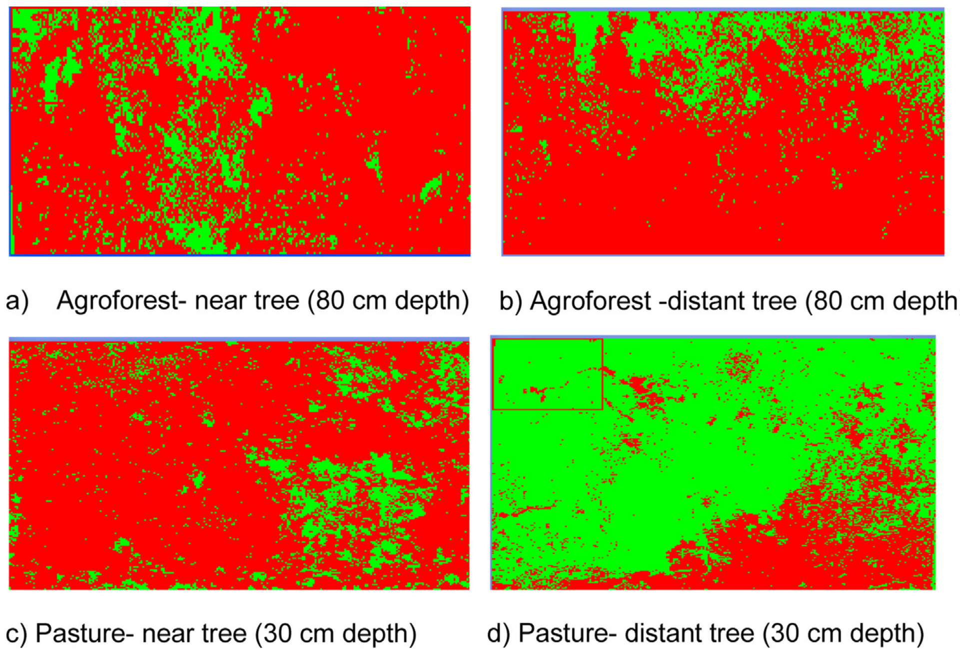

Fig. 2. Examples of stained areas (white) of pit walls at: (a) Aquiares at a location 1.9 m away from a tree and beneath its canopy, stained area of 25%; (b) Aquiares at a site 18.3 m from the closest tree, stained area of 28%; (c) Copan, beneath a clump of trees, stained area of 9%; and (d) open-pastures in Copan, 12 m from the closest clump of trees, stained area of 85%. (*for printing version*).
Examples of stained areas (green) of pit walls at: (a) Aquiares at a location 1.9 m away from a tree and beneath its canopy, stained area of 25%; (b) Aquiares at a site 18.3 m from the closest tree, stained area of 28%; (c) Copan, beneath a clump of trees, stained area of 9%; and (d) open-pastures in Copan, 12 m from the closest clump of trees, stained area of 85%. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

accounted for less than 50% of the total precipitation during all rainfall events with the exception of one that took place in May.

There is no pronounced dry season in Aquiares. However, some months have higher levels of intense rainfall. Rainfall events of more than 50 mm h^{-1} accounted for 27% (101 mm) of the rainfall in January. There is a marked dry season in Copan starting in September, whereas the two months with the highest rainfall intensities were May and June. Rainfall events above 50 mm h^{-1}

Table 2
Definitions of the preferential flow parameters meaning and methods for their calculation. Adapted from van Schaik (2009).

Preferential flow parameter	Calculation
Uniform infiltration depth (UID)	Depth at which the stained area strongly decreases. Indicates the depth to which matrix flow is prevalent. Here a threshold at 80% of stained areas to define UID
Maximum depth of blue stains (MD)	Maximum penetration depth of water in the preferential flow paths. Here we took the total depth of our pit, which varies according to the last stained layer visualized in the vertical walls
Total stained area (TSA)	The total stained area is an estimate of the total matrix flow. A low number generally is taken to indicate high preferential flow
Preferential flow fraction (PFF)	The fraction of the total infiltration, which flows through the preferential flow paths. This is calculated as: PFF = 1 − (UID × PW/TSA), where PFF (−); UID (cm), which is multiplied by the profile width (PW), and divided by TSA (cm^2). PW: profile width, in our case study 1 = 20 cm, and 30 cm in our case study 2
TSA/MD	Besides the above four parameters suggested by van Schaik (2009), we added the ratio between total stained area and maximum depth, in order to standardize for varying pit depth

Table 3
Measured rainfall intensities during downpour events in the Turrialba coffee agroforestry system and the Copan pasture system.

Site/event date	Total rainfall in the event (mm)	Total time of the event (min)	Time with intensity >50 mm h^{-1} (min)	% of event rainfall with intensity >50 mm h^{-1}
Turrialba–Aquiares				
6/01	43	40	20	79
11/01	35.2	60	20	82
11/01	43	60	30	88
12/09	28	49	20	85
Copan–Honduras				
18/05	48.2	40	35	93
6/02	33.5	60	15	38
26/06	29.5	60	15	48
3/07	28.8	60	10	45
14/08	27	60	10	32
20/04	31.3	60	10	30
11/05	26.5	60	10	37

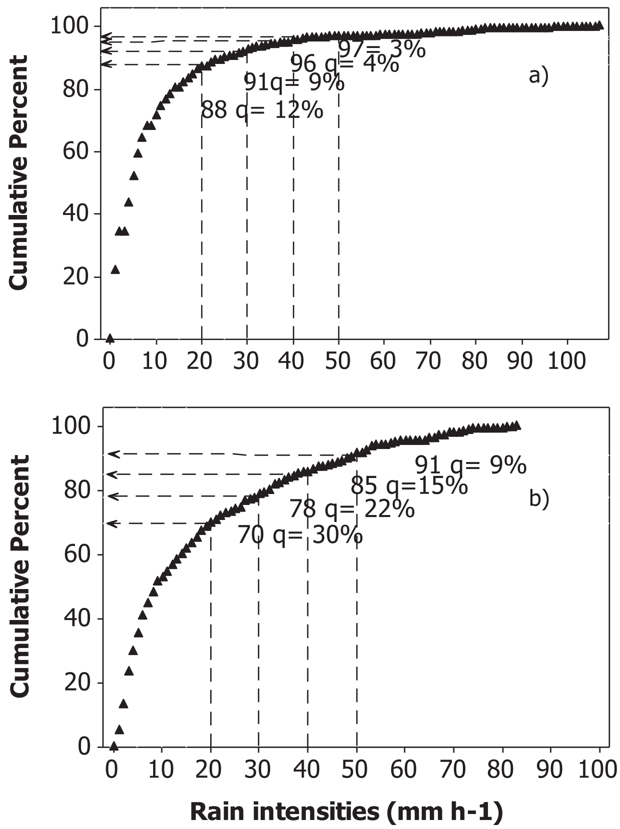

Fig. 3. Cumulative frequency distribution of rain intensities expressed in: (a) mm h^{-1} in 10-min intervals for the coffee agroforestry system over the entire study period (33 months), and (b) mm h^{-1} in 5-min intervals for the pastoral system over a period of 20 months. Dashed arrows indicate the reference rain intensities of 20, 30, 40, and 50 mm h^{-1} and the corresponding percentiles of the total rainfall.

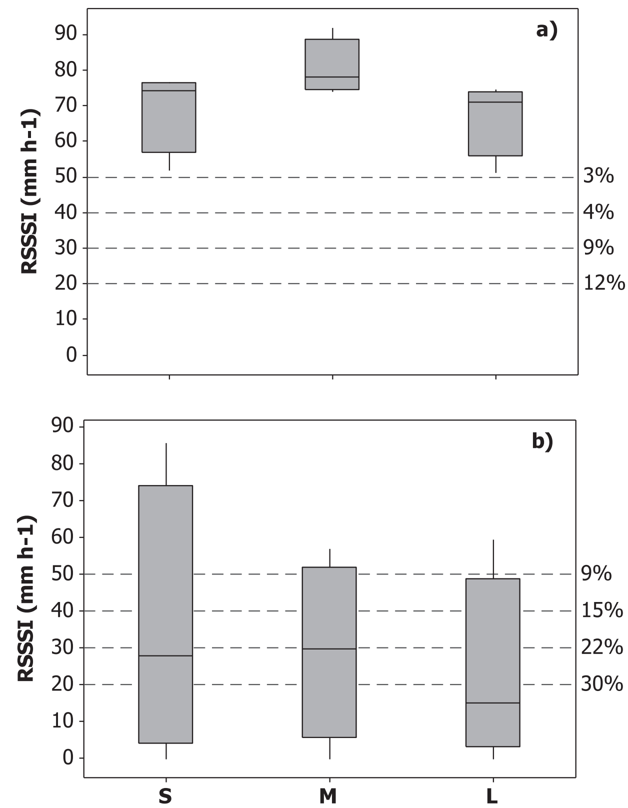

Fig. 4. Box plots showing steady state infiltrability rates (as measured by rainfall simulations—RSSSI) at L, M, and S positions (Large, Moderate, and Small distances to the nearest tree) for (a) the coffee agroforestry system (L = 22–33 m, M = 7.5–19.5 m, and S = 2.3–5 m), and (b) the pastoral system (L = 22–30 m, M = 12–15 m, and S = 1.2–2.6 m). Horizontal dashed lines show reference rain intensities and the corresponding percentiles of the rainfall distribution.

accounted for 21% (117 mm) of the total rainfall in May, and represented 6% (49 mm) of the total rainfall in June.

3.1.2. Comparison of rainfall intensities with rainfall simulation steady state infiltrability (RSSSI)

We compared the rainfall measurements to the results of the infiltration rates (as measured by RSSSI) because the simulated rainfall events more closely resembled natural rainfall.

Both the infiltration rates as measured by double ring steady state infiltrability (DRSSI) and rainfall simulation steady state infiltrability (RSSSI) indicate that there was substantial spatial variation in infiltrability. For data collected in the same general area of each transect and with the same distance to the closest tree, there was no linear relationship between the results of the double ring tests and rainfall simulations conducted at Copan, and only a weak relationship for the Aquiare's results (r^2 = 0.34, p = 0.04). The infiltrability rates (measured by RSSSI) obtained at the coffee agroforestry system, were always above 50 mm h^{-1}.The lowest measured infiltrability rates were exceeded only by a few intensive rainfall events, which accounted for only 3% of the total rainfall during our study period (Fig. 4a).

Soils in Copan have variable infiltrability rates (as measured by the RSSSI experiment: the measured levels of infiltration ranged from near-zero to well above any possible natural rainfall intensity). Copan' soils had infiltration rates that were high enough to accommodate all lower rainfall intensities, but 30% of the typical rainfall for this region was higher than approximately 25% of the measured infiltration rates (less than or equal to 20 mm h^{-1}). In addition, 66% and 75% of the measured infiltrability rates were less than or equal to 40 and 50 mm h^{-1}, respectively (Fig. 4b).

3.2. Double ring steady state infiltrability (DRSSI)

At Aquiares, there was no difference between sites in the vicinity of trees and those located further away from trees in terms of their measured infiltrability rates by double ring steady state infiltrability (DRSSI values, Fig. 5a; p = 0.105; distant tree mean= 1124 mm h^{-1}, near tree mean = 898 mm h^{-1}). In Copan (Fig. 5b), the measured infiltrability rates (by DRSSI) at sites close to trees were significantly greater than those at open-pasture sites (Mood median test, p = 0.009; clump of trees median = 146 mm h^{-1}, open-pasture median = 47 mm h^{-1}). The infiltrability rates (as measured by DRSSI) for these two site types different by an order of magnitude, strongly suggests that the infiltrability of the soils differ in real and important ways, offering a broad confirmation of the impact of trees on infiltrability.

The combination of extensive tree cover and long established coffee plants created high levels of infiltrability across the Aquiares study site (Fig. 6). Conversely, in Copan, higher infiltrability rates (>600 mm h^{-1} as measured by DRSSI) values occur exclusively at sites with nearby (<5 m) trees. Rainfall intensities of 50 mm h^{-1} are common at this site and were used as a threshold value. Infiltrability rates exceeded this threshold in 39 locations near trees while they exceeded it only in six locations far from trees.

3.3. Relationships between preferential flow variables, macroporosity, and steady state infiltrability rates

Images obtained at Aquiares, have similar levels of staining, indicating that while there is some level of preferential flow beneath

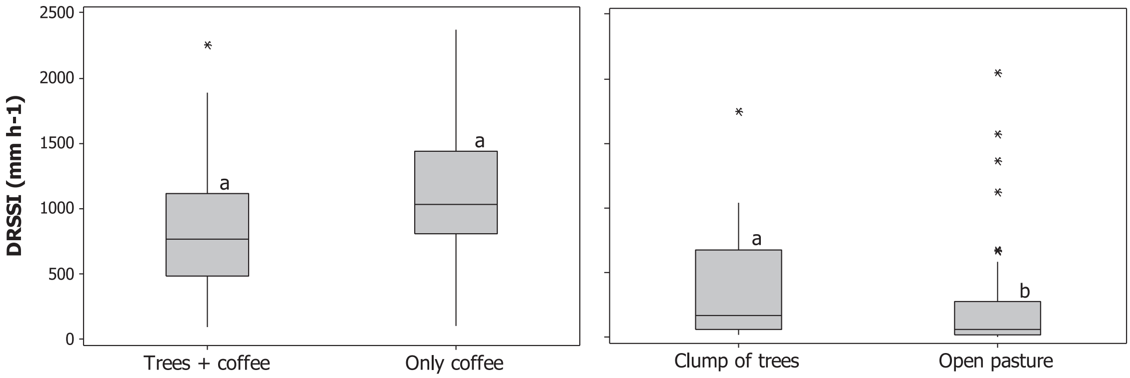

Fig. 5. Comparison of measured double ring steady state infiltrability (DRSSI) values at positions in the vicinity of trees and widely separated from trees in transects of the coffee agroforestry system (a) and the pastoral system (b), with interquartile ranges shown in boxes: top line = third quartile (75% of the measured values were equal to or less than this value), middle line = median measured value, and bottom line = first quartile (25% of the measured values were equal to or less than this value). The error bars show the maximum and minimum measured values (with outliers excluded), and asterisks indicate outliers. Outliers identified in these plots were not excluded from subsequent analyses. Boxes with different letters indicate distributions that differ significantly.

the tree (Fig. 2a), there is also good flow in the coffee-only section of the transect (Fig. 2b).

In Copan, it was found that the level of staining in the vicinity of the trees (Fig. 2c) was lower than that at sites that were distant from trees (Fig. 2d) for any given depth. This indicates that there is a greater level of preferential flow in the vicinity of the trees than is found elsewhere. The shapes of the stained pores in the open-pasture sections of the transects were consistent with matrix flow.

In the coffee agroforestry system, there was a positive correlation between total stained area/maximum depth of blue stain (TSA/MD) and distance-to-tree (as explained by PC1, Table 4). Additionally, there was a positive correlation between macroporosity at 50 cm (MC50), preferential flow fraction (PFF), and steady state infiltrability rates (RSSSI; as explained by PC2, Table 4).

With the exception of total stained area/maximum depth of blue stain (TSA/MD), no significant relationships between the distance-to-tree (DTT) variable and the various preferential flow indexes were identified for the coffee agroforestry system (Fig. 7). However, multiple relationships of this sort were identified for the pasture system in Copan.

The trees situated in the pasture landscape of Copan had positive effects on macroporosity, preferential flows, and infiltrability. Specifically, for this site, the distance-to-tree variable correlated significantly with the measured steady state infiltrability rates as well as the various preferential flow indexes. Total stained area (TSA), total stained area/maximum depth of blue stain (TSA/MD), maximum depth of blue stain (MD), uniform infiltration depth (UID), maximum porosity at 50 cm (MC50), and steady state infiltrability rates (RSSSI) where all negatively correlated to distance-to-tree (as explained by PC1, Fig. 8). Infiltrability was more closely associated with macroporosity at a depth of 50 cm than with macroporosity at 0–10 cm.

Furthermore, in the pastoral system in Copan there was a negative correlation between preferential flow fraction (PFF) and distance-to-trees (DTT), meaning that the larger the distance to trees, the lower the preferential flow, and vice versa. Overall, in this pasture system we can see the association between macroporosity and infiltrability, but also, that preferential flow fraction (PFF) and maximum depth of blue stain (MD) are negatively correlated with uniform infiltration depth (UID) and distance-to-tree (DTT).

4. Discussion

4.1. Effect of trees and of distance-to-trees on infiltrability and preferential flow

For the pasture system at Copan, we were able to confirm the general positive effect of trees on infiltrability (Ilstedt et al., 2007; Hassler et al., 2011). At Copan, the magnitude of the trees' effects

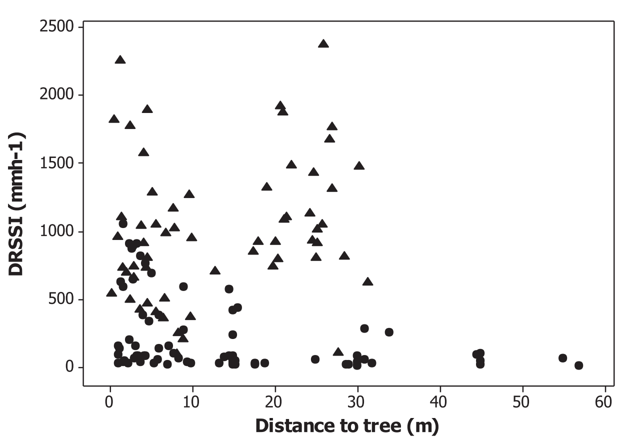

Fig. 6. The relationship between the distance to the nearest tree and the variation in double ring steady state infiltrability (DRSSI) values in the coffee agroforestry system (triangles) and the pastoral system (circles).

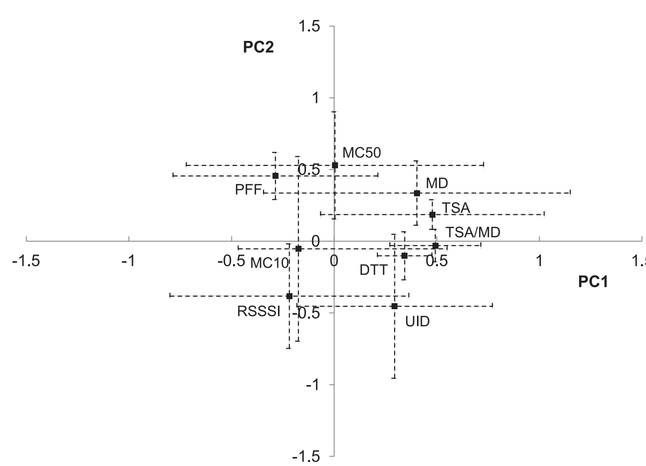

Fig. 7. Loadings and standard errors at the 95% confidence level for the first two principal components (PC) relating preferential flow parameters, distance-to-tree, infiltrability rates (as measured by rainfall simulation-RSSSI), and macroporosity for Aquiares (variable acronyms are given in Table 4).

Table 4
Variation explained (R^2) by the first two principal components, in terms of the preferential flow parameters, the distance-to-tree (DTT) parameter, steady state infiltrability rates, and macroporosity (MC) in the transects for Aquiares and Copan.

Variable	Aquiares—R^2		Copan—R^2	
	First principal component	Second principal component	First principal component	Second principal component
TSA/MD[a]	0.93	0.002	0.79	0.04
TSA[b]	0.88	0.07	0.75	0.15
MD[c]	0.62	0.24	0.61	0.27
DTT[d]	0.51	0.02	0.35	0.12
UID[e]	0.37	0.42	0.50	0.40
PFF[f]	0.35	0.42	0.20	0.65
RSSSI[g]	0.20	0.29	0.75	0.07
MC10[h]	0.13	0.006	0.40	0.04
MC50[i]	0	0.56	0.67	0.07

[a] TSA/MD: Total stained area/maximum depth of blue stain.
[b] TSA: Total stained area.
[c] MD: Maximum depth of blue stain.
[d] DTT: Distance-to-tree.
[e] UID: Uniform infiltration depth.
[f] PFF: Preferential flow fraction.
[g] RSSSI: Rainfall simulation steady state infiltrability.
[h] MC10: Macroporosity at 10 cm depth.
[i] MC50: Macroporosity at 50 cm depth.

on infiltrability depended heavily on the distance to the nearest clump of trees. This is consistent with previous reports of elevated infiltration levels under scattered tree canopies in open grasslands in Kenya (Belsky et al., 1993) and in tree belts in Australian pastures (Ellis et al., 2006). In our case, it was important to determine the tree-effect on infiltrability as a function of the distance-to-tree, since this would make it possible to model infiltration as a function of tree density. However, the trees at the agroforestry system did not appear to have any effect on infiltrability. This may be due to (1) soil type, (2) presence of perennial vegetation, and (3) no tillage. The high permeability is proper of the volcanic soils at Aquiares (Cattan et al., 2009; Dorel et al., 2000; Gómez-Delgado et al., 2011; Nanzyo, 2002). The composition of the understory, consisted of tall perennial woody coffee plants (high mean of 2 m), with extensive root systems, which were pruned regularly with the residues being left on the top soil (Defrenet, 2012), and produce large quantities of leaf residues (Beer, 1988). This agroforest is characterized by the absence of tillage and near-complete absence of soil compaction (which only occurs beneath access trails). Overall, the combination

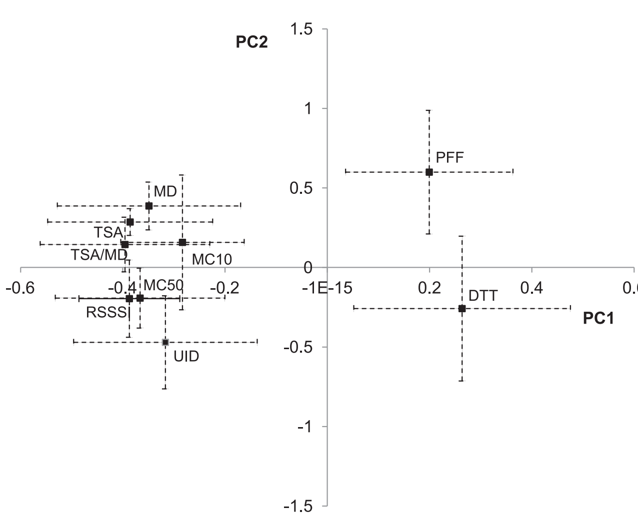

Fig. 8. Loadings and 95% confidence intervals for first two principal components (PC) for preferential flow parameters, distance-to-tree, macroporosity, and infiltrability rates (as measured by rainfall simulation-RSSSI) for Copan (variable acronyms are given in Table 4).

of these factors means that the soil's macroporosity is sufficiently high even without shade trees that the additional effect of those trees is negligible. Well managed coffee plantations are probably as beneficial for the top soil and macroporosity as are trees. However, other studies conducted on Costa Rican Andisols have shown that coffee agroforestry systems can have higher infiltration rates than coffee monocultures (Cannavo et al., 2011). In Aquiares, the contribution made by *E. poeppigiana* (biomass and nutrient return) to the soil's organic matter content and structure (Beer, 1988; Payan et al., 2009), may be less pronounced than those reported for *Inga densiflora* by Cannavo et al. (2011). In addition, Aquiares differs from the site studied by Cannavo et al. (2011) in that it has not pronounced dry season and lower annual rainfall intensities.

There was no strong linear correlation between our infiltrability rates measured by double ring and rainfall simulations (DRSSI and RSSSI). However, the same trends were apparent in both datasets with regards to the effects of trees. Touma and Albergel (1992) found that the variation in the results obtained using double ring infiltrometer is much greater than that achieved with rainfall simulators. This is partly because infiltrability values obtained using the double ring method is not limited by the water supply, unlike the case for rainfall simulations. Another reason is the high spatial variation in soil structure, together with the fact that the two methods are used to take measurements in the same general area but not in exactly the same location. Double ring measurements are also conducted on smaller overall areas of soil which increases their variability. This study did not aim to resolve these methodological problems, which will have to be addressed in order to facilitate detailed hydrological landscape modeling. Instead, we used both methods to enhance our understanding of the roles of trees in the two contrasting systems. In this sense, the two methods are complementary and the results obtained in each case are corroborative since they show similar trends regarding the impact of trees at the two study sites.

Macroporosity is well known to improve infiltrability and soil hydraulic conductivity (Bouma, 1981; Hillel, 1998). The effects of trees on macroporosity, preferential flow and infiltrability were markedly different in the two study sites. At Aquiares, such correlations were insignificant or weak. At Copan, preferential flow was dominant close to trees while matrix flow increased with distance to trees and was prevalent at sites located more than 15 m from the nearest tree. It is likely that this occurs because (1) tree roots swell, shrink, die and decompose, all of which promote macropore formation, and (2) trees add organic matter to the soil via pruning and deposition of residues in the soil, and root turnover. This will tend to increase soil microbe activity and the abundance of fauna such as termites and earth worms, all of which increase macroporosity and soil aggregation, and thus have a positive effect on soil infiltration.

4.2. The relationship between infiltrability and rainfall intensity

The beneficial effects of trees on infiltrability in the pasture system persisted when rainfall intensity was considered. The infiltrability values measured in the pasture sites varied significantly, but many (25%) of the infiltrability rates (as measured by RSSSI) observed at pasture locations were lower than the rainfall intensity of 25% of the rainfall events observed during the study period. This would be expected to cause substantial surface runoff, as was observed during rainfall simulations. These findings are consistent with those of Hassler et al. (2011), who observed differences in topsoil K_{sat} (soil hydraulic conductivity at saturation) values following pasture restoration (involving the introduction of trees under secondary succession). This was also beneficial for infiltration. The high levels of infiltrability observed for the agroforestry system are in agreement with a simulation of water flux partitioning performed at the watershed-scale in the Turrialba basin using a lumped hydrological model (Gómez-Delgado et al., 2011), in which it was predicted that the infiltration rates can accommodate 92% of the region's rainfall. In contrast to the situation at the pasture site, the infiltrability rates (measured by RSSSI) at Aquiares were always greater than the intensities of the rainfall events that occurred during the study period. In such cases, overland flows are rare: only 3% of the total rainfall observed during the course of this work was sufficient to probably cause their creation. Moreover, during the rainfall simulation steady state infiltrability (RSSSI) tests, the observed runoff volumes were negligible (no more than a few mm) after more than 2 h of simulated rainfall. Agroforestry on Andisols is thus very effective conserving water and preventing erosion (Gómez-Delgado et al., 2011).

4.3. The importance of inherent and prevailing conditions (soil, land use, understory)

It can reasonably be assumed that the soils in the pastures of Copan will be less permeable than would those of natural forests in the same region. While no direct comparison was performed in this work, this finding is not surprising due to roughly 20 years of trampling, erosion, and grazing in these pastures. In such cases, tree roots systems and litterfall would be expected to significantly improve the physical properties of the soil (Ilstedt et al., 2007; Scott et al., 2005). Conversely, the coffee agroforestry system was converted form forest roughly 40 years ago, but has maintained perennial vegetation ever since. The plantation's coffee bushes are highly productive woody plants with deep and extensive root systems, and are regularly pruned (with the residues being left on the topsoil). This is likely to have maintained the favorable functioning and physical properties of these soils. Notably, the Andisol soils at Aquiares may be more stable in terms of their physical properties than the Argiustolls at Copan, which are more readily degraded due to changes in land use. The degradation profile of the soil is thus an important inherent factor that has a significant effect on the potential co-beneficial impact of introducing trees in order to improve soil physical properties. Other important factors, outside of the scope of this study, include the identity of the introduced tree species (Malmer et al., 2010), the method used to establish the trees, and a range of edaphic variables (Ilstedt et al., 2007).

Many studies have investigated the theoretical and methodological aspects of preferential flow (Jarvis, 2007). However, few have studied the relationships between land use and management, preferential flow, and infiltrability. Work has shown differences between sites containing trees and open-pasture, for example, the wetting fronts under acacia canopies were deeper than those under open grasslands in West Kenya (Belsky et al., 1993). A study in Hawaii found higher preferential flow in areas containing native tree species (individually and in various combinations) compared to nearby degraded grassland (Perkins et al., 2012).

Preferential flow and macroporosity has been shown to be altered by land management in many different climates, farming practices and landscape management (Joffre and Rambal, 1988; Tarrá et al., 2010; van Schaik, 2009).

Our findings regarding the positive effects of trees on infiltrability are also relevant in terms of the reported increase in heavy precipitation events for Central America (Aguilar et al., 2005), and the likely increase in runoff in areas where grasslands are becoming more common and rates of evapotranspiration are reducing. Such scenarios are commonly anticipated as a consequence of climate change, as discussed by Aguilar et al. (2005) and Imbach et al. (2010). Such changes may make pasture systems increasingly prone to superficial runoff and the attendant site erosion, while also promoting the flooding of areas at lower altitudes. The adoption of agroforestry systems should provide some protection against these expected climatic changes.

5. Conclusions

Measurable tree-effects on hydrological services are evident in low density environments like pastures where the accompanying species are shallow rooted, whereas, tree-effects are masked in systems with a strong presence of perennial vegetation, but the combination of tree cover and old coffee plants contribute to infiltrability an order of magnitude higher than in the pasture across the entire site. In coffee agroforestry systems, the evaluation of trees must consider the other environmental services they provide. These include enhanced fertility, higher soil organic matter, the presence of soil-stabilizing rooting systems, promotion of biodiversity, microclimate, and carbon sequestration. The results presented herein confirm the need for more wide-ranging empirical data in order to accurately predict the beneficial effects of trees on hydrological services in different environments.

Acknowledgments

We would like to thank to Hacienda Aquiares for field access, the Barquero family and Thaise S. Souza for their support in field campaigns in Aquiares; to Josue León and Karen Banegas for their support in Copan. Our gratitude for logistics and meteorological instrumentation to the Coffee-Flux Observatory which has been supported by the EU-CAFNET project, SOERE F-ORE-T network of observatories and ANR-Ecosfix project. We acknowledge the Projects MESOTERRA from CATIE; Trees, Carbon & Water from SLU, and the Schlumberger Foundation Faculty for the Future, for their financial support.

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